
Software Requirements Specification (SRS): Canteen Inventory & Sales System

1. System Overview

A lightweight, automated inventory and sales management system designed specifically for canteen operations. It replaces manual paper ledgers with automated tracking of raw materials, recipe-based stock deduction, daily sales, and official reporting.

2. User Roles

- **Canteen Manager / Operator:** Enters daily purchases, updates daily meal preparation counts, and logs sales/wastage.
- **System Admin:** Manages master data (recipes, prices, users) and configures stock alert limits.
- **Officer (Read-Only/Signatory):** Reviews generated daily reports for sign-off.

3. Core Functional Requirements (Feature List)

3.1. Master Data Management (The Foundation)

Before the system can automate tasks, it needs basic setup features.

- **Item Master:** Add/edit raw materials (Fresh, Dry, Packaging) with their units of measurement (Kg, Ltr, Pcs).
- **Menu/Product Master:** Define the final items sold (e.g., Standard Lunch, VIP Thali, Tea).
- **Pricing Management:** Set Cost Price (CP) for raw materials and Selling Price (SP) for final meals.

3.2. Auto-Stock Deduction (Recipe System)

- **Recipe Creator:** Define exact ingredient quantities required for one unit of a finished meal (e.g., 1 Lunch = 100g Rice, 50g Dal, 1 Box).
- **Batch Preparation Entry:** Input the total number of meals prepared for the day/shift.
- **Automated Inventory Deduction:** The system automatically calculates total ingredients used based on the batch size and subtracts them from the main inventory.
- **Wastage Logging:** Feature to record unsold or spoiled food, ensuring accurate profit/loss calculations.

3.3. Daily Digital Ledger (In & Out Tracking)

- **Goods Received Entry:** Log newly purchased stock with quantity and total cost.
- **Automated Balance Calculation:** System automatically maintains the formula:
 $\text{Opening Stock} + \text{Received Stock} - \text{Issued (Cooked) Stock} = \text{Closing Stock}$.
- **Category Filtering:** View ledgers separately for Dry Rations, Fresh Vegetables, Dairy, and Packaging materials.
- **Manual Adjustment (Audit):** Admin-only feature to manually adjust stock in case of physical counts not matching system counts (records a reason for the adjustment).

3.4. Daily Sales & Profit Tracker

- **Sales Entry:** Input total meals sold for the day.
- **Revenue Calculation:** Automatically multiplies items sold by their Selling Price (SP).
- **Cost of Goods Sold (COGS) Calculation:** Automatically calculates the total cost of raw materials consumed that day.
- **Real-time Profit Display:** Dashboard widget showing: $\text{Total Revenue} - \text{Total Cost of Ingredients} = \text{Net Daily Profit}$.
- **Payment Mode Tracking:** Tag sales revenue as Cash, UPI, or Card for accurate bank deposit tracking.

3.5. Smart Inventory & Alerts

- **Reorder Level Configuration:** Set a "minimum stock limit" for every individual item (e.g., Sugar: 2 Kg).
- **Dashboard Alerts:** Color-coded visual warnings (e.g., Red) on the home screen for items that have dropped below their minimum limit.
- **Shopping List Generator:** One-click generation of a purchase list based on low-stock items.

3.6. One-Click Daily Reports & Sign-offs

- **Daily Summary Generation:** Auto-compiles Opening Balance, Purchases, Usage, Closing Balance, Sales, and Cash collected into a single, standardized view.
- **Print-Ready Formatting:** Generates a PDF optimized for A4 printing, complete with structured tables.
- **Signature Blocks:** Auto-generates empty signature rows at the bottom of the report (e.g., "Prepared By", "Checked By", "Approved By JCO/Officer").
- **Historical Report Access:** A calendar view to pull up the signed report for any previous date.

4. Non-Functional Requirements (Performance & UI)

Category	Requirement

Usability	The interface must be simple enough for non-technical canteen staff to use with minimal training. Large buttons and clear fonts are required.
Accessibility	Mobile/Tablet responsive. Canteen staff should be able to enter daily prep numbers via a tablet in the kitchen.
Security	Password-protected logins. Regular operators cannot change historical data without Admin approval.
Data Backup	Daily automated database backups to prevent loss of financial and stock records.